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COMSOL Multiphysics 6.2 Documentation

Model Wizard:

This is the first step to create the model. First selection is about **Space Dimension**. It is usually selected as a 2D model or 3D model. After that **Physics Selection** page is opened. Motor design models usually need **Magnetic Field (mf)** or **Rotating Machinery, Magnetics (rmm)**.

Magnetic Field: The Magnetic Fields interface is used to compute magnetic field and induced current distributions in and around coils, conductors, and magnets. Depending on the licensed products, stationary, frequency-domain, small-signal analysis and time-domain modeling are supported in 2D and 3D.

Rotating Machinery, Magnetics: The Rotating Machinery, Magnetic interface is used for design and analysis of electric motors and generators. Stationary and time-domain modeling is supported in 2D and 3D. The physics interface solves Maxwell's equations formulated using a combination of magnetic vector potential and magnetic scalar potential as the dependent variables.

Final step is to choose the **Study**. Stationary and Time-Dependent studies are available. In case of use of coil in the design, **Coil Geometry Analysis** should be added and solved before any other study.

Global Definitions:

Under **Global Definitions** ( ) you add features that apply to the entire model. Add the following features by right-clicking **Global Definitions** and choosing an option from the context menu. You can also access the most widely used functionality using the toolbars.

Parameters are user-defined global, scalar values that are used to parameterize any part of the model. It makes easy to change design parameter for the entire model. In Figure 1, there is an example for parameters definition. Under the **Expression** section, it is important to write the correct unit for the variable.

Variables node can be used to define expressions as user-defined ***variables***. Global variables can be used in any context that accepts variable expressions, in all Components and on all geometric entities provided that their expressions are also global expressions.

Function templates for creating user-defined functions based on analytic expressions or imported data, or specifying parameters for common function types such as step functions, ramps, and random functions. In Figure 2, all available functions are shown.

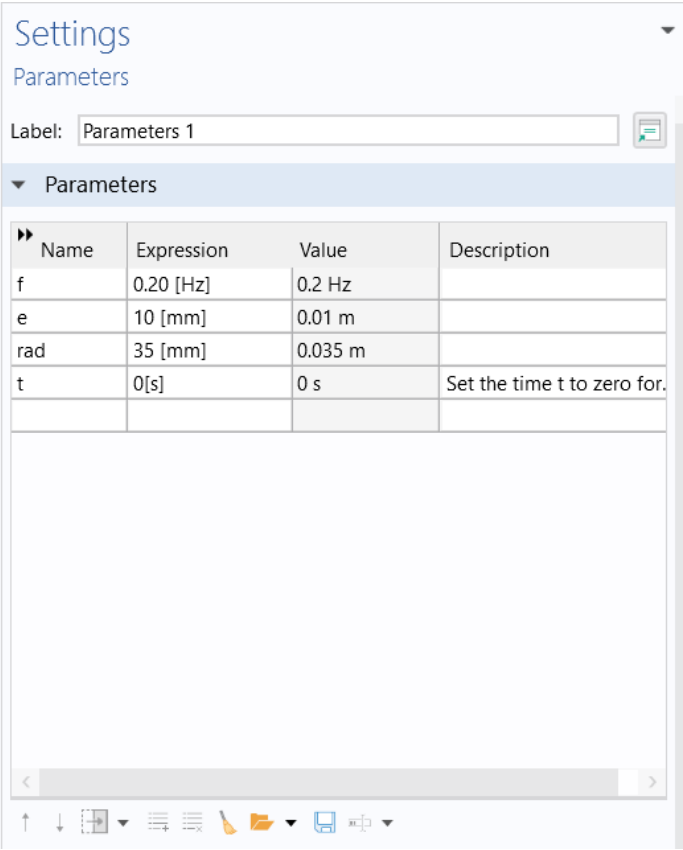

Figure 1: Parameters

Figure 2: Functions

Component:

A model component is a fundamental part of the model and contains a geometry with its associated physics interface, mesh, and variables and other definitions that are local to that component. The **Component** node defines the namespace for each part of a model that is defined in a model component.

In component settings one can choose the unit system, frame and curved mesh elements. In **Curved Mesh Elements**, geometry shape function can be chosen according to the model. By default, the software avoids inverted elements by an optimization of the element shapes. To deactivate that functionality, clear the **Avoid inverted elements by curving interior domain elements** checkbox. Inverted mesh can be a serious warning or an error. For more information, please see the link provided in **Useful Links** section.

Definitions:

In this subsection **Identity Boundary Pair** can be seen. Use an identity pair node (for an identity boundary pair) to specify two selections of boundaries that overlap but belong to different parts of an assembly. Then assign a boundary condition to connect the physics nodes in the two parts in a physics interface. Identity pairs connect overlapping boundaries in different connecting parts of an assembly. This is important to apply physics in different domains. Identity boundary pair can be created by using **Form an Assembly** option. It is mentioned in detailed in the next section. For more explanation, please see the Useful Links section.

In a motor design it is important to separate the rotor and stator domains. Using unions and other operations, create a single geometry object for the stationary part and another one for the rotating part, and apply the physics accordingly.

Different coordinate systems such as cylindrical or spherical coordinate systems also can be added by right clicking on the ***Definitions*** -> ***Coordinate Systems***.

Geometry:

In this subsection **Length Unit** should be chosen before building the geometry.

There are different ways to generate the geometry for the design and simulations.

1. Draw the geometry within the COMSOL® software
2. Import an external CAD file
3. Import mesh data from an external file
4. Choose an object from the selection of built-in geometric shapes in the software.

The built-in geometric shapes are located in **Part Libraries**, right click on Geometry node then it is under Parts node. Figure 3 shows some types of built-in geometries.

The **Geometry** branch in the Model Builder tree consists of a sequence of geometry operations. This can include sketches, geometric primitives, imported CAD files, Boolean operations, etc., that can be used to create the geometry that you want to model.

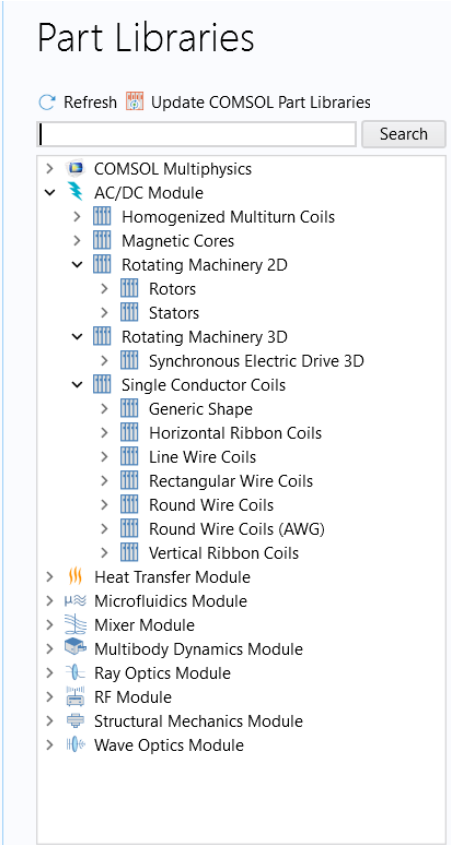

Figure 3: Part Libraries

After generating a few objects in the geometry sequence, you can start to combine them in meaningful ways using geometry operations. This could be done by using a combination of several Boolean and transformation operations.

At the end of this sequence, there is a ***Form Union*** node, and you have the option to change this to ***Form Assembly***. This choice of method influences how the final geometric model used for the physics settings and meshing is generated.

Form Union: A single geometry object is created and composed of many different domains. Any overlapping geometry objects will become a single object composed of different domains for each enclosed volume. In the physics settings, each of these different domains can have different material properties. By default, continuity of the fields and the fluxes will be maintained across the interior boundaries, and it is possible to add a variety of different discontinuity or jump conditions at interior boundaries for the various physics.

Form Assembly: It differs from *Form Union* in that it does not compute a Boolean union of the geometry objects. Instead, it collects them into an assembly object. After implementing the *Form Assembly* operation, adjacent objects will become disconnected domains and movement is possible if needed. Because of this, you need to switch to *Form Assembly* for the following physics interfaces when you have touching objects in your geometry sequence:

- *Solid Mechanics* interface, when *Contact* features are included
- *Multibody Dynamics* interface
- *Rotating Machinery, Magnetic* interface
- *Rotating Machinery, Fluid Flow* interfaces

Since *Form Assembly* does not compute the Boolean union of the objects, you cannot use *Form Assembly* to generate domains for the overlapping regions between objects. The number of domains in the geometry will not change as a consequence of the *Form Assembly* operation. For detailed explanation please see the Useful Links section.

Materials:

The Material Library product contains a comprehensive set of material properties, all of which have references. In Figure 4, the material library can be seen.

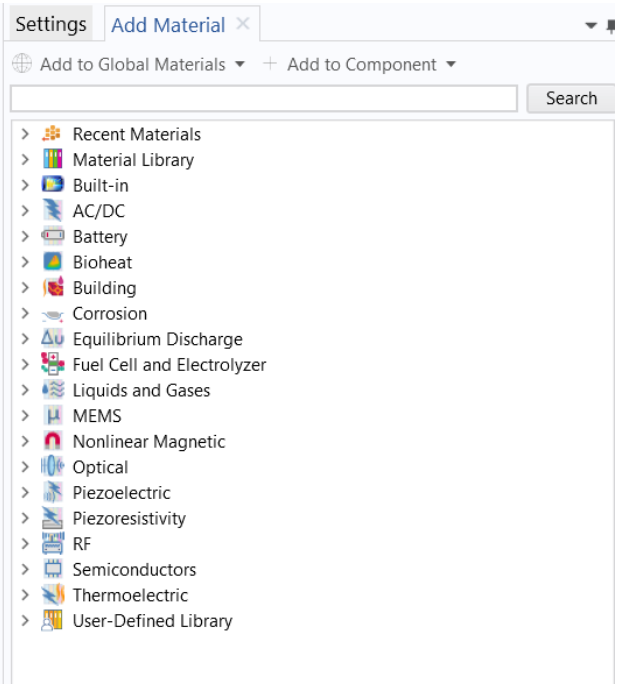

Figure 4: Material Library

After the material selection, domains can be selected for the related material. In the settings window of a material, there is a **Material Contents** node, shown in Figure 5. From here value of some properties can be changed. As it can been in the Figure 5, there are some boxes checked next to some properties. That means this material is used in some physics with those properties. If you right click on a property, you can click on **Go to Requested Node** to see where that property is used.

Material Contents

Property	Variable	Value	Unit	Pr
Relative permeability	mur_is...	1	1	Ba
Relative permittivity	epsilo...	1	1	Ba
Electrical conductivity	sigma...	10[S/m]	S/m	Ba
Coefficient of thermal expansi...	alpha_...	alpha_p...	1/K	Ba
Mean molar mass	Mn	0.02897[...	kg/mol	Ba
Bulk viscosity	muB	muB(T)	Pa·s	Ba
Dynamic viscosity	mu	eta(T)	Pa·s	Ba
Ratio of specific heats	gamma	1.4	1	Ba
Heat capacity at constant pres...	Cp	Cp(T)	J/(kg·K)	Ba
Density	rho	rho(pA,T)	kg/m³	Ba

Figure 5: Material Contents

Moving Mesh:

They can be used to study both stationary states and time-dependent deformations where the geometry changes its shape due to motion of solid boundaries and deformation of solid domains. The **Moving Mesh** interface can be added by navigating to the **Home** tab and selecting it from the **Physics** section. It can be added by right clicking on **Component** as well. Rotating domain, Prescribed Deformation, and Deforming Domain are used in our designs so far.

Rotating Domain: Add a Rotating Domain node ( )to add rotation to the selected domains. You can specify the rotation type and the rotation axis as can be seen from the Figure 6.

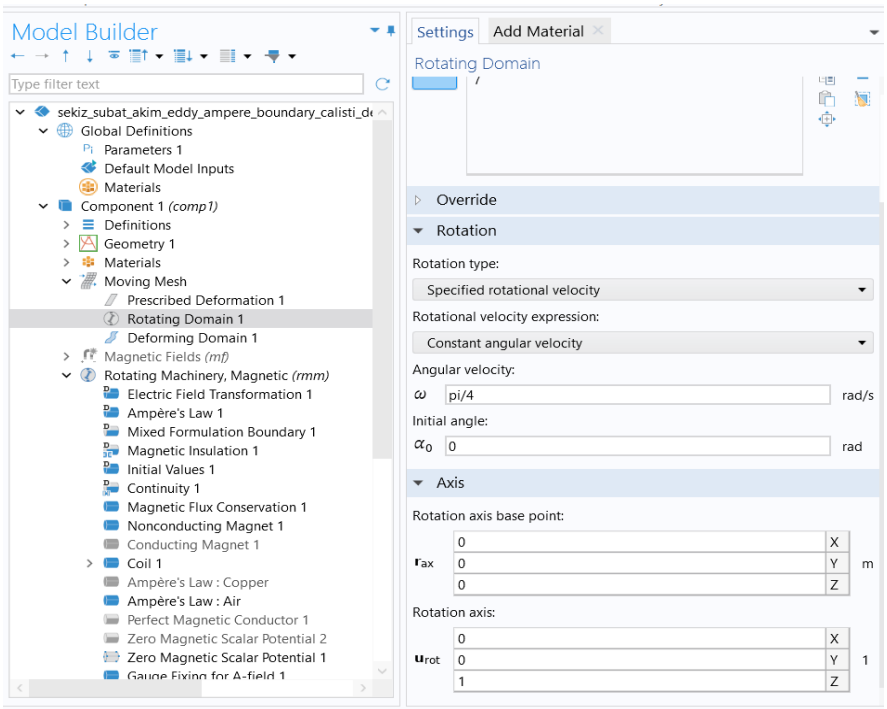

Figure 6: Rotating Domain Settings

From the Rotation Type list user defined, specified rotational velocity and rigid body options can be chosen. According to the selection you can set the rotation properties.

Rotor domains should be chosen under Rotating Domain selection to define the rotation.

Prescribed Deformation: Use the Prescribed Deformation node ( ) to explicitly define the deformation of the mesh in one frame relative to another. Note that the Solid Mechanics interface has built-in behavior corresponding a Prescribed Deformation node under Moving Mesh.

This property can be used to model the wobble movement or to specify an angle of rotation (which can be time dependent) or use the *Prescribed Rotational Velocity* feature to enter a constant angular velocity.

It is important that the stator's initial position is set to the point where the movement will start. Otherwise, a significant stress is created on meshes. To reduce the stress on the meshes one should create union between air and stator. However, it does not necessary for rotor and air.

Deforming Domain: Add a Deforming Domain node ( ) to specify that the shape of the selected domains should be governed by the domain boundaries, which are in turn controlled either by explicit deformed mesh boundary condition nodes, or by implicit continuity constraints requiring that a deforming domain must follow the shape of adjacent domains, or stay fixed if there is no adjacent domain. Explicit boundary conditions take precedence over implicit constraints.

In the setting page, Smoothing node is important. For the mesh smoothing, choose smoothing type from the **Mesh smoothing type** list: *Laplace*, *Winslow*, *Hyperelastic*, or *Yeoh* (the default). For Yeoh, also enter a Stiffening factor C_2 (default: 10). For detailed explanation please see the Useful Links section.

Deforming Domain should be defined to domains which are adjacent to the domains selected for Prescribed Deformation.

Physics:

In this part we mainly focused on Rotating Machinery, Magnetic (rmm) but there are common properties with Magnetic Fields (mf).

Rotating Machinery, Magnetics (rmm):

In any rotating magnetic machine, there are two parts: the *stator* and the *rotor*, separated by an air gap enabling the rotor's rotation. The *Rotating Machinery, Magnetic* interface uses the moving mesh approach to model this rotation, as the finite element method does not support rotations.

The *Rotating Machinery, Magnetic* interface solves Maxwell's equations to compute the distribution of the electromagnetic field. Most quantities of interest (e.g., the applied torque) can be computed once the fields are known. In time-dependent analyses, the interface applies the quasistatic approximation, which neglects the displacement current density, or equivalently assumes that capacitive effects in the machine are negligible. With this approximation, all of the currents in the machine are either externally applied (i.e., through an excited winding) or are eddy currents induced in the machine's conductive parts. The nonconductive parts, like the air gap, do not carry any current density.

There are two approaches used in this interface to solve Maxwell's equations: the *vector potential formulation* and the *scalar potential formulation*. In the former approach, a vector field, A (the magnetic vector potential), is introduced and the approach defines the magnetic flux density and the electric field as

$$E = -\frac{\partial A}{\partial t}$$

$$B = \nabla \, x \, A$$

With these definitions, the B and E fields automatically fulfill two of Maxwell's equations:

Faraday's law and the magnetic flux conservation law (or magnetic Gauss' law). These are written as such:

$$\nabla \, x \, E = -\frac{\partial B}{\partial t}$$

$$\nabla B = 0$$

The equation to be solved is Ampère's law:

$$\nabla \, x \, H = J$$

The vector potential formulation is used in the *Magnetic Fields* physics interface. The scalar potential formulation is only applicable in regions where the electric current density is zero. In this case, a scalar field V_m (the magnetic scalar potential, not to be confused with the electric potential) is introduced and the magnetic field is defined by the approach as the gradient of this potential. With this definition, Ampère's law is automatically fulfilled and the magnetic flux conservation law is solved. This formulation is used in the *Magnetic Fields, No Currents* physics interface.

Compared to the vector potential formulation, the scalar potential formulation introduces fewer degrees of freedom and leads to an "easier" problem to solve. The downside is, of course, that it can only be used in the absence of currents. Normally, this condition would restrict the applicability to special cases, like stationary studies of permanent magnets. But, thanks to the quasistatic approximation, this formulation can also be applied to nonconductive regions in time-dependent analyses.

These two formulations can also be used together by combining the vector potential formulation for conductive or current-carrying domains and the scalar potential formulation for the air gap and nonconductive domains. Referred to as ***mixed formulation***, this approach is particularly useful in 3D models due to an increased accuracy of the pair coupling given by the scalar formulation. In 2D models, for in-plane magnetic fields, the discretization scheme used for the vector potential is similar to that used for the scalar potential. Thus, in 2D in-plane cases, using the mixed formulation is not necessary.

In the setting page of Rotating Machinery, Magnetic click on *Discretization* to choose discretization of MVP and MSP.

Some physics are mentioned below.

Electric Field Transformation:

The **Electric Field Transformation** node imposes suitable transformations to the electric field definitions in all domains depending on rotational velocity. The default property cannot be modified.

Ampère's Law:

The **Ampère's Law** node adds Ampère's law for the magnetic field to a domain and provides an interface for defining the constitutive relation and its associated properties as well as electric properties.

In the Settings page, click on Constitutive Relation B-H to choose Magnetization Model, and Relative Permeability. Relative Permeability, B-H Curve, Remanent Flux Density, and Non-linear Permanent Magnet options are commonly used as Magnetization Model.

The Ampère's Law is used in domains where the magnetic vector potential (MVP) formulation is used. Meaning that in conducting domains.

Mixed Formulation Boundary:

The **Mixed Formulation Boundary** node imposes continuity between the magnetic scalar potential on one side of the boundary and the magnetic vector potential on the other side. In other words, on one side the Ampère's Law is applied and on the other side the Magnetic Flux Conservation is applied. This feature is added by default and is applied automatically to all appropriate boundaries.

Magnetic Insulation:

The **Magnetic Insulation** node is the default boundary condition for the Magnetic Fields interface and adds a boundary condition that sets the tangential components of the magnetic potential to zero at the boundary $n \times A = 0$.

Initial Values:

The **Initial Values** node adds initial values for the mass fractions that can serve as an initial condition for a transient simulation, or as an initial guess for a nonlinear solver. If required, add additional Initial Values nodes from the Physics toolbar. In our designs, Magnetic Vector Potential and Magnetic Scalar Potential are set to 0 by default.

Continuity:

Add a **Continuity** node to connect parts of an assembly using one or several **Identity boundary pairs**. Both sides of the identity pair must be part of the physics interface in which you add the **Continuity** node.

This is important to apply physics to domains thar are separated by Identity Boundary Pair and to solve the model correctly. For detailed information on Constraint Settings please see the link given in Useful Links.

Magnetic Flux Conservation:

Magnetic Flux Conservation provides an interface for defining the constitutive relation and the relevant material properties (for example, the relative permeability).

Click on Constitutive Relation B-H in the setting page to see Magnetization Model and Relative Permeability. Select **B-H curve** to use a curve that relates magnetic flux density B and the magnetic field H as $|\mathbf{B}| = f(|\mathbf{H}|)$. To define $|\mathbf{B}|$, select *From material* (the default) to use the value from the material or select *User defined* to specify a value or expression for the magnitude of the magnetic flux density in the field that appears.

The Magnetic Flux Conservation is used in domains where the magnetic scalar potential (MSP) formulation is used. Meaning that in non-conducting domains. The MSP formulation introduces fewer degrees of freedom and ensures a more accurate coupling of the magnetic flux density when used with a pair feature like *Continuity* or *Sector Symmetry*. Therefore, it is important to solve an air region on either side of the identity pair through the MSP.

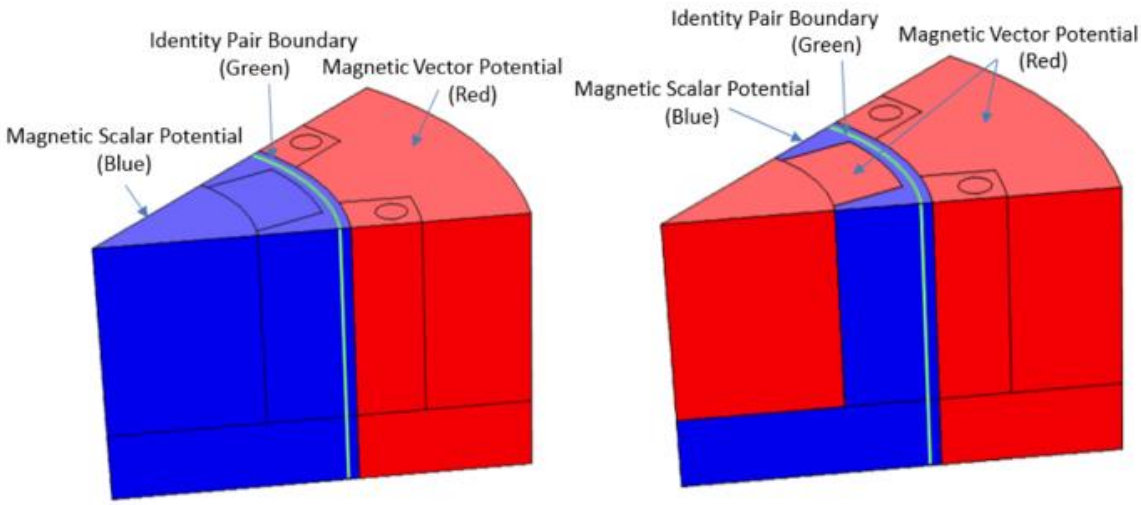

Figure 7: The MSP and MVP regions, along with an identity pair boundary.

In the Figure 7 on the left, the stator coil, stator iron, and air region on the stator side are modeled with MVP. The rotor iron, permanent magnet, and air region of the rotor, as well as a part of the stator's air region, are modeled with MSP. Using this setup, you can model the entire MVP region with the *Ampère's Law* feature. Since the MVP formulation allows for current-carrying domains, it is possible to model the stator iron region for the induced eddy currents and compute the eddy current losses. The same, however, cannot be said for the rotor iron, as it is an MSP formulation and explicitly assumes that there is no current in the regions. If you wish to compute the eddy current losses both on the stator and rotor iron (or if you have any other conducting domains in the rotor), you must also solve for those regions with MVP, as illustrated in the above figure on the right. For detailed explanation please see the link given in Useful Links section.

Conducting & Non-Conducting Magnet:

Click on *Magnet* in the *Settings* page to view Direction Method. If you use *User defined* option you can choose *Direction of Magnetization*. If you choose *Specify north and south boundaries* option, you can use pattern type options.

Zero Magnetic Scalar Potential:

The Zero Magnetic Scalar Potential node provides a boundary condition that specifies a zero magnetic potential on the boundary $V_m = 0$. The selection of zero point may not be critical but it should be chosen if MSP is applied in a geometry.

Gauge Fixing for A-field:

The Gauge Fixing for A-Field node enforces the gauge $\nabla \cdot \mathbf{A} = 0$ by adding an additional potential variable, ψ, and its associated conservation equation to the system. This is often necessary to get a unique and numerically stable solution to the equation solving for the magnetic vector potential A.

In 3D, there is an option to change the feature's *Equation form*, which is set to *Study controlled* by default. For Time Dependent and Frequency Domain studies, set Equation form to *Stationary* can be efficient to handle regions without conductivity.

The variable ψ is used to impose a condition on the derivatives of the magnetic vector potential, so its absolute value does not have particular significance; only its gradient enters the equations (the variable ψ acts more or less like a potential). The absolute value of the variable can be set by entering the *Divergence condition variable scaling* ψ_0 (SI unit: A/m). The default value is 1 A/m, which is appropriate for most models.

Gauge fixing is usually required when the solution of the $\mathbf{A}$ field is not unique and you want to use a direct solver. Using a direct solver may be required or beneficial in cases involving multiphysics couplings, as iterative solvers and preconditioners typically are tuned for one specific type of physics only. Also, when solving 2D models, direct solvers are used as default.

It is recommended not to use gauge fixing unless necessary, since it increases the total number of unknowns, requiring more memory and time to solve. Depending on different models, the computation time could increase by 30–60% if gauge fixing is used. It is also worth mentioning that the gauge fixing will result in the system matrix having zeros on the diagonal. The latter is not a problem when using a direct solver, but an iterative solver has to use more expensive methods in the preconditioner. For detailed information please see the link in the Useful Links section.

Coil:

The Coil feature supports two different *Conductor models*:

Single Conductor, which models a conductive body such as a wire, busbar, or other metallic object in which the current flows freely due to the material's conductivity. This

model can be used when the current flow has a well-defined beginning and end or is a closed loop.

Homogenized multiturn, which models a bundle of electrically thin wires tightly wound together but separated by an electrical insulator. In this scenario, the current flows only in the direction of the wires. Induced current flowing in other directions are considered negligible. Skin and proximity effects can be included in a lumped sense, using the Harmonic loss effective loss model or a Mutually coupled circuit.

There are 3 different coil types.

Numeric Coil Types: In a Numeric coil, the current flow is computed automatically in a Coil Geometry Analysis study step.

Circular Coil Types: In a Circular coil, the wires are wound in circles around the same axis. Use the Coil Geometry sub-node to select a group of edges forming a circle or a part of a circle around the coil's axis. From the selected edges, the coil axis is computed and the direction of the wires is taken to be the azimuthal direction around the axis, as marked by the red arrows. The coil length used is computed as the coil volume divided by the coil cross sectional area, unless the **Use robust geometry analysis method** box is checked. When the robust method is used, the coil length is simply the length of the selected edges.

Linear Coil Types: In a Linear coil, all the wires are parallel and straight lines. Use the Coil Geometry sub-node to select an edge or a single group of connected edges that maps out the local coil direction. The direction of the wires and the coil length is taken to be the direction and the length of the edge(s), as marked by the red arrow. Avoid selecting multiple parallel edge groups as that will result in an incorrect coil length.

To respect the current conservation law, the applied currents cannot originate from interior boundaries. A Linear coil should therefore be terminated on exterior boundaries.

For detailed information please see the link in the Useful Links section.

Meshes:

Usually, physics-controlled mesh can be used in designs. If a simulation takes a large amount of computational time to solve then user-controlled mesh may be required. In 3d models it is better to use Free Tetrahedral. In the settings page of Free Tetrahedral, click on Element Quality Optimization, and check the *Avoid inverted curved elements* box.

Study:

Coil Geometry Analysis:

The Coil Geometry Analysis study is used to compute the current flow of a Coil feature in 3D models. This is a preprocessing step that must be solved before solving the main study step. The best results are obtained when the coil has a smoothly varying or constant cross section without sharp bends and bottlenecks. The local current directions are solved for by the specialized Coil Geometry Analysis study step.

Stationary:

Stationary solution can be used to calculate B-field, losses and torque values.

Time Dependent:

In electromagnetics, it is used to compute transient electromagnetic fields. You do not need to consider the settings in the Time-Dependent Solver unless the simulation does not behave as expected. For example, the simulation behaves in an unexpected way when changing the end time, in case you may need to adjust the default settings for the time stepping and the tolerances. You may also need to output more time steps and use a strict time stepping and tolerance to fully resolve fast dynamics and to avoid aliasing effects.

Select a Time unit from the list (default: s) to use a time unit that is convenient for the time span of the simulation. Then specify the time interval for the output from the simulation in the Times field using the selected time unit. You can type a monotonically increasing list of individual values, for example, 0 1 2 5 10 20; use the range operator, for example, range(0,0.1,1.5), which gives time steps from 0 to 1.5 s with a step size of 0.1 s; or use any combinations of such input.

Select the **Automatic remeshing** checkbox if you want the solver to remesh automatically when the quality of the mesh becomes poor in a Time Dependent study. Select the geometry to use for the automatic remeshing from the Remesh in geometry list. With automatic remeshing active, the solver adds an Automatic Remeshing sub-node under the Time-Dependent Solver node. In that sub-node, you specify the mesh quality expression that determines when to remesh.

Useful Links:

- Avoiding Inverted Mesh Elements: https://doc.comsol.com/6.3/doc/com.comsol.help.comsol/comsol_ref_mesh.24.28.html#534979
- Identity Boundary Pair: https://doc.comsol.com/6.3/doc/com.comsol.help.comsol/comsol_ref_definitions.21.118.html
- Form Assembly: https://www.comsol.com/support/learning-center/article/the-usage-of-form-union-and-form-assembly-74571
- Smoothing Methods: https://doc.comsol.com/6.3/doc/com.comsol.help.comsol/comsol_ref_deformedmeshes.34.09.html#723774
- Rotating Machinery Tutorial: https://www.comsol.com/model/download/1170991/models.acdc.rotating_machinery_3d_tutorial.pdf
- Constraint Settings: https://doc.comsol.com/6.3/doc/com.comsol.help.sme/sme_ug_solid.07.128.html
- Use of MSP & MVP in modeling of Rotating Machines https://www.comsol.com/blogs/guidelines-for-modeling-rotating-machines-in-3d
- Gauge Fixing https://www.comsol.com/blogs/how-do-i-use-gauge-fixing-in-comsol-multiphysics
- Modeling Electromagnetic Coils https://www.comsol.com/support/learning-center/course/modeling-electromagnetic-coils-112/modeling-electromagnetic-coils-8251
- Coil https://doc.comsol.com/6.3/doc/com.comsol.help.acdc/acdc_ug_magnetic_fields.08.057.html
- Studies and Solvers p1031 https://doc.comsol.com/5.3a/doc/com.comsol.help.comsol/COMSOL_ReferenceManual.pdf